

Design and Implementation Plan for a Web-Based Multiplayer Nation Roleplay Game

Executive summary

The strongest direction is **not** to build “NationStates with combat,” and it is also **not** to build a full Paradox-style spreadsheet simulator in the browser. The best product fit is a **hybrid persistent nation RPG / grand-strategy sandbox** that borrows NationStates’ policy-event readability and nation identity tools, Crusader Kings III’s character drama and agent logic, Hearts of Iron IV’s mission-driven armies and templates, Europa Universalis IV’s province-and-trade abstraction, Civilization VI’s accessible map readability and governor-like specialization, and web-nation-sim patterns from Politics & War, eRepublik, and Supremacy-style browser games. NationStates’ official FAQ makes clear that its core loop is broad policy choice via recurring Issues, not micromanagement, and that nations cannot directly attack or destroy one another; that is precisely why it is a great template for **identity, writing, and diplomacy**, but a poor template for the user’s desired **map war game** unless heavily reworked. ¹

The recommended product architecture is a **web-native TypeScript stack: React + TypeScript + Vite** for the application shell, **PixiJS** for the 2D map renderer, **NestJS** on **Fastify** for HTTP/API services, **Colyseus** for authoritative realtime state sync in map rooms and battles, **PostgreSQL + Prisma** for persistent data, **Tiled JSON + a custom province graph** for map authoring/runtime, and **Cloudflare R2 + Images + Stream** for user media. This stack is the best fit because the product is half strategy game and half content platform: it needs both rich CRUD-style UI and a performant tactical map. The official docs support each of those components’ strengths: Vite as modern web tooling, PixiJS as a high-performance 2D web renderer, NestJS as a scalable TypeScript backend, Fastify as a low-overhead web framework, Colyseus as an authoritative multiplayer framework with state synchronization, PostgreSQL for transactions and JSON/JSONB, Prisma for type-safe data access and migrations, and Cloudflare’s storage/media products for S3-compatible storage plus image/video delivery. ²

The design should intentionally **de-emphasize regions** as the social center. NationStates puts substantial weight on regions, delegates, endorsements, and region message boards, but the user explicitly does not want region governance to dominate. Instead, the social graph should pivot around **nation profiles, embassies, friends/followers, bilateral treaties, coalition memberships, and public/private nation feeds**, while still allowing optional “cultural regions” or map continents for flavor. NationStates dispatches already show the right separation of **factbook-style canon, bulletins, and story accounts**, and eRepublik’s newspaper system demonstrates that player-made news can become a durable communication layer. ³

The recommended MVP is therefore: **persistent nations + policy events + province map + cities/ports/mines/bases + customizable unit templates + army orders + character agents with XP/assignments + nation news/media feed + friendships/treaties**. The **family/life-sim layer** for the player avatar should be deliberately postponed until after the core nation/war/agent loop is fun and stable. CK3 proves that deep character simulation can carry an entire grand strategy game, but its breadth is also a warning against trying to ship the full dynasty/life-sim stack too early. ⁴

Lessons from NationStates

NationStates succeeds because it gives players a nation identity quickly, then repeatedly asks for meaningful, readable decisions. Its official FAQ says players develop their nations by answering Issues that appear several times a day, and that the game emphasizes broad policy decisions rather than tweaking fine-grained numbers. That is the single most valuable NationStates lesson to adopt. The event layer in the new game should therefore be the primary identity engine: short, high-legibility decisions that alter laws, culture, unrest, diplomacy, resources, and character reputations over time. ⁵

NationStates also demonstrates the value of **nation-as-document**. Players can create Dispatches under clearly separated types—**Factbooks, Bulletins, Accounts, and Meta**—with rich formatting, links, and images; community guidance further distinguishes objective nation information from in-universe stories and region-board discussion. That should be adopted almost directly: every nation in the new game should have an **official profile**, a **newsroom**, and a **role-play archive**, each with different editorial defaults and visibility settings. ⁶

Where NationStates is weakest for this project is warfare and physical world simulation. Its FAQ explicitly says nations cannot be attacked or destroyed by foreign powers, and its “military” gameplay is displaced into regional endorsement politics, delegate power, and role-play forums such as International Incidents. That model is useful as a warning: it produces a very strong writing/diplomacy community, but it leaves military play largely external to systemic nation gameplay. Because the user wants cities, bases, ports, mining sites, military units on a 2D map, and character-led armies, the new game must treat warfare as **first-class gameplay**, not as forum theater. ⁷

NationStates’ region system should be **partially adopted, structurally demoted**. Regions have message boards, founders/governors, delegates, influence, and invasion politics, but the user does not want design effort dominated by region administration. The correct translation is to keep only the useful parts: **opt-in blocs, embassies, friends networks, coalition halls**, and maybe **continent-wide public boards**. Do not make “regional office politics” the core progression path. ⁵

A concise priority matrix follows.

Priority	NationStates mechanic	Adopt or avoid	Why
Highest	Recurring Issues with broad policy outcomes ⁵	Adopt	This is the cleanest identity engine and the best onboarding loop.
Highest	Nation profile + factbook/dispatch culture ⁶	Adopt	Perfect fit for nation RP, canon, lore, and “official state media.”
High	Telegrams / direct outreach and forum discussion ⁷	Adopt in lighter form	Necessary for diplomacy, role-play invites, and coalition politics.
Medium	World Assembly-style global law arena ⁷	Adopt later	Valuable for endgame diplomacy, but not needed for MVP fun.

Priority	NationStates mechanic	Adopt or avoid	Why
Low	Region governance, endorsements, delegates, founder powers ⁵	Minimize	Interesting, but directly conflicts with the user's priorities.
Avoid	No direct warfare between nations ⁵	Avoid	The user wants visible armies, positional strategy, and military missions.

Comparative system analysis

The target design space becomes clearer when the core systems are compared directly across the games and services most relevant to this project.

Game or system	Event complexity	Agent depth	Military model	Map type	Multiplayer model	Modability assessment	Best design lessons
NationStates ⁸	Frequent, short, policy-text events	Low at character level; nation is the primary "actor"	No direct nation warfare; region politics and RP substitute for combat	No tactical war map	Persistent browser world	Content-level only via dispatches, role-play, issue submissions; no true mod SDK visible	Build through concise choices; player authority canon
Civilization VI ⁹	Medium; eras, loyalty, governors, climate/emergency-style systems	Medium; governors and leaders shape play, but not deep personal drama	Unit-based empire warfare with districts and map expansion	Hex map	Single-player plus online play and teams	High ; official mod tools + Steam Workshop	Use re-terrain special and appropriate military pacing
Crusader Kings III ¹⁰	Very high; thousands of dynamic, character-tailored events	Very high ; traits, stress, lifestyles, schemes, guardianship	Men-at-arms, knights, levies, mercenaries, commander-driven wars	Province/county grand-strategy map	Primarily campaign multiplayer and single-player	High ; Steam Workshop and strong scripting culture	Character-driven strategy just do it

Game or system	Event complexity	Agent depth	Military model	Map type	Multiplayer model	Modability assessment	Best design lesson
Hearts of Iron IV ¹¹	High; focus trees and historical/alternate-history progression	Low-to-medium; leaders matter less than state apparatus	Division templates, battle plans, logistics, fronts, up to 32-player MP	Global province map	Competitive/co-op multiplayer	High ; Steam Workshop and mod community	Keep planning system mission-oriented twitch
Europa Universalis IV ¹²	High; hundreds of historical events, ruler effects, trade and diplomacy shifts	Medium; rulers/advisors matter, but less personally than CK3	Army/navy on province map with trade and diplomacy as equal systems	Province world map	Up to 32-player cross-platform multiplayer	High ; Steam Workshop and mod support	Make diplomacy trade, war pay one step simula
Politics & War ¹³	Medium; browser-nation management with wars, alliances, cities, evolving narrative	Low; nation page is primary identity surface	Browser nation war with alliance politics and cities	Nation/city-oriented browser presentation	Persistent browser MMO	Low-to-medium ; strong player lore/API feeling, but no deep official mod stack visible	Brows nation can su long-li alliance politic layere nation
eRepublik ¹⁴	Medium; politics/economy/war/world events	Medium at citizen level; citizens can own businesses, vote, write, fight	Regional war campaigns between countries and allies	World countries + regions	Persistent MMO social strategy	Low-to-medium ; player-created newspapers and politics, but limited visible engine modding	Player media citizen can m world lived-i

Game or system	Event complexity	Agent depth	Military model	Map type	Multiplayer model	Modability assessment	Best design lessons
Supremacy 1914 ¹⁵	Low-to-medium; seasonal/live-content structure rather than deep narrative events	Low-to-medium; commanders/generals are tactical enhancers	Real-time province warfare with attrition, diplomacy, logistics, huge player counts	Large persistent map, up to 500 players	Long-running realtime browser campaigns	Low in official tooling terms	Large, burnable maps sustain meaningful diplomacy and territorial struggles

The system synthesis is straightforward. For **events**, NationStates, CK3, and EU4 are the most relevant inspirations: NationStates for short policy choices, CK3 for character-scoped drama, and EU4 for geopolitical randomness and historical texture. For **characters**, CK3 is the gold standard, while Civ VI's governors show how to keep role assignment readable and low-friction. For **military**, HOI4 is the best source for battle plans, templates, and mission-driven war, while Civ VI is the best source for map readability and unit comprehensibility. For **nation media**, NationStates dispatches and eRepublik newspapers are the clearest models to copy. ¹⁶

The design implication is that the new game should be positioned as:

Nation identity and writing tools from NationStates + character agency from CK3 + strategic armies from HOI4/Civ + browser persistence from web nation sims.

That combination fits the user's requirements much better than trying to clone any single predecessor. ¹⁷

Recommended technical stack

Stack options and tradeoffs

Layer	Strong option	Advantages	Risks	Codex-friendliness	Recommendation
App shell	React + TypeScript + Vite ¹⁸	Fast local dev, simple SPA structure, easy integration with custom canvas map	You assemble routing/data patterns yourself	Excellent because file structure, typing, and boundaries are explicit	Use

Layer	Strong option	Advantages	Risks	Codex-friendliness	Recommendation
App shell	Next.js ¹⁹	Powerful React framework with full-stack features	Adds routing/runtime conventions and extra server complexity not strictly needed here	Good	Use only if you want SSR-heavy public pages very early
2D map renderer	Pixijs ²⁰	High-performance 2D rendering on WebGL/WebGL2/WebGPU/Canvas; ideal for layered map sprites and overlays	Requires you to build game interaction patterns yourself	Excellent because it stays close to the browser and TypeScript	Use
2D game framework	Phaser ²¹	Mature HTML5 game framework, strong scene loop	More opinionated for “game screens,” less natural for dashboard-heavy nation sim UI	Good	Viable alternative if the product becomes more “game window” than “web platform”
Browser engine export	Godot Web export ²²	Real engine workflow, strong 2D tooling	Web export constraints; weaker fit for feed/forms/admin surfaces	Fair	Not recommended for the primary client
Browser engine export	Unity Web/ WebGL ²³	Mature tooling and asset pipeline	Heavier web client; WebGPU experimental; more friction for content-heavy browser UI	Fair	Not recommended for the primary client

Layer	Strong option	Advantages	Risks	Codex-friendliness	Recommendation
Backend HTTP/API	NestJS on Fastify ²⁴	Structured TypeScript backend, modular architecture, Fastify performance option	Slightly more ceremony than a minimal Node server	Excellent for generated/refactored code	Use
Realtime game state	Colyseus ²⁵	Authoritative rooms, schema-based state sync, matchmaking	Another abstraction to learn	Excellent because room state is explicit and typed	Use
Realtime notifications	Socket.IO ²⁶	Low-latency event messaging with fallbacks	Less domain-specific than Colyseus for authoritative game rooms	Good	Optional adjunct, not the core simulation layer
Primary database	PostgreSQL ²⁷	Transactions for nation economy/battles, JSONB for flexible event payloads	Requires careful schema discipline	Excellent when paired with Prisma	Use
Data access and migrations	Prisma ORM ²⁸	Type-safe client, migrations, strong TypeScript ergonomics	Some advanced SQL features still need raw SQL or custom migrations	Excellent	Use
Map authoring	Tiled JSON + custom province graph ²⁹	Easy authoring, custom properties, layered art, good browser portability	Needs a runtime importer and validation tooling	Excellent because the content is declarative	Use

Layer	Strong option	Advantages	Risks	Codex-friendliness	Recommendation
Image/ media storage	Cloudflare R2 + Images + Stream 30	S3-compatible object storage, image transformations, unified video pipeline	Cloudflare-centric operational footprint	Excellent via presigned/direct-upload workflows	Use
Alternative media stack	Amazon S3 + Cloudinary/Mux 31	Industry-standard object storage and very strong media tooling	More vendors and more integration points	Good	Good fallback if Cloudflare-first deployment is not desired

Final recommended stack

The final recommendation is:

- **Client:** React + TypeScript + Vite
- **2D map renderer:** PixiJS
- **Backend API:** NestJS with Fastify adapter
- **Realtime simulation:** Colyseus rooms for map sectors, battles, and presence-critical world state
- **Persistence:** PostgreSQL + Prisma
- **Map authoring:** Tiled JSON for art layers, plus a versioned custom province-graph JSON for gameplay
- **Media:** Cloudflare R2 for files, Cloudflare Images for image delivery, Cloudflare Stream for video
- **Content definitions:** JSON/YAML event templates with schema validation
- **Repository shape:** pnpm monorepo with shared types between client, server, and content packages

This is the right choice because the product is fundamentally a **content-heavy strategy application**, not a pure game-engine executable. Nation pages, feeds, diplomacy forms, media uploads, moderation tools, admin editors, and lore archives are all first-class surfaces. A web-native stack will make those surfaces much easier to ship than a browser-exported engine client, while PixiJS still gives the performance needed for an attractive 2D strategic map. 32

Codex-followable repository layout

```
repo/
  apps/
    web/
      src/
        app/
        screens/
        components/
        map/
```

```
    features/  
    api/  
api/  
  src/  
    modules/  
      auth/  
      worlds/  
      nations/  
      diplomacy/  
      posts/  
      media/  
      events/  
      characters/  
      armies/  
      battles/  
      colyseus/  
      jobs/  
packages/  
  shared/  
    src/  
      types/  
      enums/  
      validators/  
      dto/  
  content/  
    events/  
    traits/  
    doctrines/  
    buildings/  
    unit-templates/  
map-data/  
  worlds/  
  tilesets/  
  province-graphs/  
config/
```

That layout is intentionally optimized for generated code, explicit contracts, and content-driven iteration. The most important principle is that **game rules live in declarative content files wherever practical**, while the server owns authoritative execution.

Detailed design specification

Core design pillars

The product should be built around five pillars:

Pillar	What it means in practice
Nation identity	Nations must feel like authored states, not just account records
Strategic readability	A player should understand the map, the war, and the consequences of a choice quickly
Character agency	Important actions should often be performed through a character, not just by clicking a nation button
Persistent role-play	Nations need durable lore, state media, speeches, and diplomatic channels
Browser accessibility	Sessions should be meaningful in 5–15 minutes, with deeper play available for long sessions

Those pillars are the synthesis of the source games: NationStates’ readable policy loop, CK3’s character-first drama, Civ VI’s visual legibility, and HOI4/EU4’s strategic state systems. ³³

Gameplay loop

The recommended loop is **hybrid asynchronous**:

- A **global daily tick** updates economy, happiness, legitimacy, construction, loyalty, resource extraction, event eligibility, and passive agent progress.
- A **short tactical tick** resolves army movement, patrols, sieges, blockades, scouting, and ongoing battles.
- Players interact continuously through UI updates and notifications, but the game is not balanced around always-on micromanagement.

That cadence preserves the “persistent nation sim” feel of browser games while still giving military operations visible positional meaning. NationStates proves that players tolerate and even enjoy a slower, periodic policy loop, while Supremacy-style games prove that long-running territorial campaigns work in the browser. ³⁴

Nation profile and media model

Every nation should have three clearly separated content surfaces:

Surface	Purpose	Default tone	Editable by
Profile	Canon facts about the nation	Official / encyclopedic	Nation owner and trusted editors
Newsroom	Press releases, updates, speeches, war bulletins	Official / current	Nation owner and authorized characters
Stories	RP accounts, civilian stories, diplomacy scenes	Narrative / in-world	Nation owner and authorized writers

This directly improves on NationStates' factbook/account split and adds a first-class media feed inspired by eRepublik newspapers and the lore-heavy nation pages visible in Politics & War. 35

Recommended profile tabs:

- Overview
- Government and Laws
- Cities and Infrastructure
- Military
- Culture and Religion
- Diplomacy
- News
- Archive
- Media Gallery

Visibility states:

- Public
- Friends
- Allies
- Private / Internal Government

Data model

World and nation entities

Entity	Key fields
users	id, email, username, password_hash, created_at, last_login_at, is_admin
player_profiles	user_id, display_name, avatar_media_id, locale, timezone, privacy_json
worlds	id, slug, name, status, tick_config_json, created_at
nations	id, world_id, owner_user_id, slug, name, adjective, government_type, ideology_social, ideology_economic, currency_name, motto, leader_title, capital_settlement_id, flag_media_id, color_hex, stability, legitimacy, influence, treasury, research_points, created_at
nation_stats_daily	nation_id, date, population, gdp_proxy, approval, unrest, food_balance, steel_balance, fuel_balance, manpower, audit_json
laws	id, nation_id, domain, key, value, enacted_at, source_event_instance_id

Entity	Key fields
relationships	id, nation_a_id, nation_b_id, relation_type, trust_score, rivalry_score, status, created_at, updated_at
treaties	id, type, proposer_nation_id, status, terms_json, starts_at, ends_at
friendships	user_id, friend_user_id, status, created_at
posts	id, nation_id, author_character_id, post_type, title, body_md, summary, visibility, cover_media_id, published_at, edited_at
post_media	post_id, media_asset_id, sort_order, caption
media_assets	id, owner_user_id, provider, kind, storage_key, mime_type, width, height, duration_sec, size_bytes, moderation_status, created_at

Map and economy entities

Entity	Key fields
provinces	id, world_id, slug, name, terrain, biome, polygon_json, center_x, center_y, owner_nation_id, controller_nation_id, population, loyalty, unrest, supply_value, passability, visibility_mask_json
province_neighbors	province_id, neighbor_province_id, movement_cost, is_sea_route, is_river_crossing
resource_sites	id, province_id, resource_type, richness, capacity_per_day, depletion, owner_nation_id, is_discovered
settlements	id, province_id, nation_id, kind, name, level, hp, population, garrison_capacity, production_queue_json, slots_json, visual_anchor_x, visual_anchor_y
structures	id, settlement_id, building_type, level, operational_status, construction_progress, queue_position
stockpiles	nation_id, money, food, timber, iron, coal, oil, steel, aluminum, munitions, rare_earths
trade_offers	id, seller_nation_id, resource_type, quantity, price_per_unit, status, expires_at

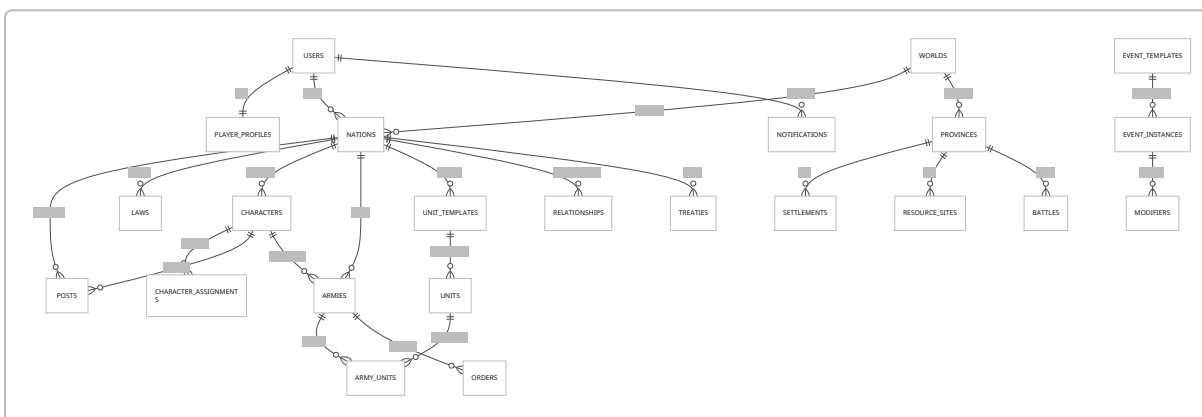
Characters, agents, and military entities

Entity	Key fields
characters	id, nation_id, is_player_avatar, first_name, last_name, title, portrait_media_id, age, health, energy, loyalty, level, xp, current_status, location_province_id, martial, diplomacy, stewardship, intrigue, charisma, biography_md
character_traits	character_id, trait_key, tier, source
character_skills	character_id, skill_key, rank
character_assignments	id, character_id, assignment_type, target_type, target_id, mission_params_json, started_at, eta_at, status
unit_templates	id, nation_id, name, branch, chassis, role, icon_key, base_stats_json, equipment_slots_json, cost_json, upkeep_json, doctrine_tags_json
units	id, nation_id, template_id, name, commander_character_id, current_hp, morale, supplies, veterancy_xp, status, location_province_id, customization_json
armies	id, nation_id, name, commander_character_id, stance, location_province_id, visibility, supply_mode, order_state
army_units	army_id, unit_id, position_order
orders	id, issuer_character_id, subject_type, subject_id, mission_type, path_json, target_province_id, status, progress_pct, issued_at, eta_at
battles	id, world_id, province_id, terrain_snapshot_json, weather_snapshot_json, started_at, ended_at, result, summary_md
battle_participants	battle_id, side, nation_id, army_id, unit_id, character_id, losses_json, xp_gained

Event and simulation entities

Entity	Key fields
event_templates	id, namespace, key, title, category, scope_type, trigger_json, weight_expr, choices_json, cooldown_hours, tags_json, is_repeatable, version
event_instances	id, template_id, world_id, scope_type, scope_id, nation_id, character_id, payload_json, status, presented_at, expires_at, resolved_at, selected_choice_key
modifiers	id, scope_type, scope_id, modifier_key, stacks, value_json, starts_at, ends_at, source_event_instance_id
notifications	id, user_id, kind, payload_json, read_at, created_at
audit_logs	id, actor_user_id, entity_type, entity_id, action, diff_json, created_at

ER diagram



API surface

Essential HTTP endpoints

Area	Method	Endpoint	Purpose
Auth	POST	/auth/register	Create account
Auth	POST	/auth/login	Start session
Auth	POST	/auth/logout	End session
Worlds	GET	/worlds	List available worlds/shards

Area	Method	Endpoint	Purpose
Worlds	GET	/worlds/:worldId/summary	Tick status, map summary, population stats
Nations	POST	/nations	Create nation
Nations	GET	/nations/:nationId	Read nation profile
Nations	PATCH	/nations/:nationId/profile	Edit identity fields, tabs, canon text
Nations	PATCH	/nations/:nationId/laws	Enact or change law values
Nations	GET	/nations/:nationId/feed	Nation news and story feed
Posts	POST	/nations/:nationId/posts	Publish update, news, speech, or story
Posts	PATCH	/posts/:postId	Edit post
Media	POST	/media/direct-upload	Get upload token / URL
Diplomacy	POST	/relationships/:nationId/friend-request	User-level friend request tied to nation
Diplomacy	POST	/treaties	Propose treaty
Diplomacy	PATCH	/treaties/:treatyId/respond	Accept/decline
Map	GET	/map/:worldId	Layer metadata and province graph
Map	GET	/provinces/:provinceId	Province details
Map	POST	/provinces/:provinceId/build	Start settlement or building project
Characters	GET	/characters/:characterId	Character sheet
Characters	POST	/characters/:characterId/assign	Assign mission or office
Characters	POST	/characters/:characterId/speech	Perform speech action at a location
Armies	POST	/unit-templates	Create or update unit template
Armies	POST	/units	Queue unit production
Armies	POST	/armies	Create army
Armies	POST	/orders	Issue move/mission order
Battles	GET	/battles/:battleId	Battle log and result

Area	Method	Endpoint	Purpose
Events	GET	/events/inbox	Pending decisions
Events	POST	/events/:eventInstanceId/resolve	Choose event outcome
Admin	PUT	/admin/event-templates/:id	Update data-driven event templates
Admin	POST	/admin/map/import	Import map package

Realtime events

The realtime surface should be narrow, typed, and authoritative.

Channel or room	Event	Payload
world:lobby	presence.updated	online nation/user counts
world:lobby	post.published	feed card metadata
world:lobby	treaty.proposed	diplomatic alert
province:{id}	province.updated	ownership, unrest, building changes
army:{id}	army.position	province movement progress
battle:{id}	battle.tick	round summary
battle:{id}	battle.resolved	winner, losses, XP
character:{id}	assignment.updated	mission state
user:{id}	notification.created	targeted alert
user:{id}	event.presented	new policy/story event

Frontend screens and components

Screen	Core components	Notes
Landing / auth	login form, world list, onboarding preview	Fast nation creation matters
Nation dashboard	nation header, quick stats, current laws, pending events, top characters, build queue	Primary “daily check-in” screen
World map	Pixi canvas, province hover cards, army sprites, route preview, strategic filters	Should be the visual centerpiece
Province sheet	owner, loyalty, structures, resources, stationed units, governor, local news	Opens from map click

Screen	Core components	Notes
Character roster	portrait cards, assignment filters, XP bars, trait chips, mission button	One of the game's signature interfaces
Character sheet	stats, traits, history, current assignment, skill tree, relationships	Deep but readable
War room	army list, unit templates, movement orders, frontline summary, battle reports	HOI-lite, not HOI-complete
Event inbox	decision cards, effects preview, involved characters, timers	NationStates-style readability
Newsroom	post composer, image/video upload, visibility selector, category tags	Dispatch/newspaper equivalent
Diplomacy	treaties, friends, allies, embassies, relationship timeline	No heavy region bureaucracy
Notifications	diplomacy, war, character, event, media moderation	Must be actionable, not noisy
Admin tools	event editor, map inspector, moderation queue, balance tables	Needed early for live-ops

Map format and render model

The best implementation is a **hybrid map model**:

- **Art layer:** Tiled JSON for terrain art, decorative layers, roads, ports, icons, and labels.
- **Gameplay layer:** custom province graph JSON for ownership, adjacency, slots, travel costs, and visible anchors on the world map.

That hybrid is the natural browser equivalent of what the source games do conceptually: Civ uses a readable tiled/hex space, while EU4/HOI4/CK3 use province graphs. Tiled's official formats already support layers, tilesets, objects, and custom properties, which makes it a good authoring source for the visual side.

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Recommended gameplay map node format

```
{
  "worldId": "alpha-world",
  "version": 3,
  "tileLayers": [
    { "name": "terrain", "source": "world-alpha-terrain.json" },
    { "name": "decor", "source": "world-alpha-decor.json" }
  ],
  "provinces": [
    {
```

```

    "id": "p_northreach",
    "name": "Northreach",
    "terrain": "plains",
    "biome": "temperate",
    "center": [312, 108],
    "polygon": [[284,82],[336,86],[348,114],[320,144],[276,126]],
    "neighbors": [
      { "provinceId": "p_ironfjord", "cost": 2, "kind": "land" },
      { "provinceId": "p_blueharbor", "cost": 3, "kind": "river" }
    ],
    "slots": [
      { "id": "s_city_1", "kind": "city", "anchor": [306,112] },
      { "id": "s_port_1", "kind": "port", "anchor": [340,126] },
      { "id": "s_mine_1", "kind": "mine", "anchor": [290,92] }
    ],
    "resources": [
      { "type": "iron", "richness": 0.82 },
      { "type": "grain", "richness": 0.34 }
    ],
    "supplyValue": 6,
    "passability": 1.0
  }
]
}

```

Sample map mockup

```

<svg width="100%" height="280" viewBox="0 0 900 280" xmlns="http://www.w3.org/2000/svg" role="img"
aria-label="Sample 2D strategic map mockup"> <rect width="900" height="280" fill="#dceaf6"/> <path
d="M40,80 C120,20 240,20 320,70 L360,130 L320,200 L200,230 L90,190 L40,140 Z" fill="#c8dba1"
stroke="#6f8f59" stroke-width="3"/> <path d="M390,55 L560,35 L680,75 L710,145 L650,210 L500,225
L410,185 L370,115 Z" fill="#c6d7a0" stroke="#6f8f59" stroke-width="3"/> <path d="M715,40 L840,60
L860,150 L810,220 L710,210 L680,130 Z" fill="#cfd8a8" stroke="#6f8f59" stroke-width="3"/> <path
d="M335,0 L370,25 L355,50 L300,60 Z" fill="#b8d6ef" /> <path d="M0,220 C100,190 180,200 250,250 L0,280
Z" fill="#b8d6ef" /> <circle cx="215" cy="120" r="12" fill="#b45f3c"/> <rect x="206" y="128" width="18"
height="8" fill="#7e4b30"/> <text x="190" y="108" font-size="14" font-family="sans-serif">Capital</text>
<circle cx="300" cy="170" r="10" fill="#8d99ae"/> <text x="276" y="158" font-size="12" font-family="sans-
serif">Town</text> <circle cx="330" cy="150" r="8" fill="#2a9d8f"/> <path d="M325,156 L338,156"
stroke="#1d6f65" stroke-width="3"/> <text x="345" y="160" font-size="12" font-family="sans-serif">Port</
text> <polygon points="470,120 490,96 510,120 490,144" fill="#6c757d"/> <text x="445" y="92" font-
size="12" font-family="sans-serif">Mine</text> <rect x="560" y="120" width="22" height="16"
fill="#495057"/> <rect x="565" y="112" width="12" height="10" fill="#6c757d"/> <text x="542" y="104" font-
size="12" font-family="sans-serif">Base</text> <circle cx="770" cy="120" r="8" fill="#e76f51"/> <circle
cx="790" cy="126" r="8" fill="#e76f51"/> <path d="M770,120 L790,126" stroke="#b23a2a" stroke-width="3"/
> <text x="742" y="104" font-size="12" font-family="sans-serif">Army</text> <path d="M224,120 C300,95
405,100 470,120" fill="none" stroke="#355070" stroke-width="3" stroke-dasharray="8 6"/> <path
d="M582,128 C650,118 714,118 770,120" fill="none" stroke="#7b2cbf" stroke-width="3" stroke-

```

```
dasharray="10 6"/> <text x="356" y="98" font-size="12" font-family="sans-serif">trade / travel route</text>
<text x="620" y="112" font-size="12" font-family="sans-serif">army order path</text> </svg>
```

Unit lifecycle

The military design should sit between Civilization’s readability and HOI4’s template logic.

Unit states

State	Meaning
designing	Template exists but is not fielded
queued	Unit production has begun
training	Unit accrues readiness and can be expedited or delayed
deployed	Unit exists on map and can be assigned
attached	Unit is assigned to an army
moving	Unit is traveling between provinces
engaged	Unit is in battle
garrisoning	Unit is defending a location
refitting	Unit is healing/re-equipping
disbanded	Unit is removed and returns partial resources or manpower

Unit customization model

Each unit template should include:

- branch: land, naval, air-later
- chassis: infantry, militia, cavalry, armor, artillery, engineer, transport, frigate, destroyer, carrier-later
- role: line, siege, scout, support, anti-armor, blockade, logistics
- equipment slots
- doctrine tags
- movement profile
- upkeep profile
- special abilities

This pulls from HOI4’s division-template logic, CK3’s men-at-arms specialization, and Civ’s accessible unit distinctions without copying their exact mechanics. 37

Character and agent lifecycle

The **character system should be a core pillar**, not a side minigame. Every nation gets:

- one **player avatar** character
- a starting cabinet of 3–5 named functionaries
- a growing roster of agents through recruitment, events, appointments, and reputation

Character roles

Role	Typical assignments
Ruler	national speeches, decrees, treaty signings, inspections
Governor	city development, unrest reduction, tax efficiency
General	army command, morale, battlefield tactics
Marshal	local defense, police action, militia training
Diplomat	embassy, treaty talks, conferences, covert overtures
Spymaster	intelligence, sabotage, secrets, counterintelligence
Steward	logistics, construction, procurement
Propagandist	public campaigns, legitimacy, war support

Character state machine

State	Description
idle	Available for assignment
traveling	Moving on the map
governing	Assigned to settlement/province
commanding	Leading army or garrison
diplomatic_mission	Negotiating or representing nation
speaking	Delivering a speech/campaign action
intrigue	Conducting covert activity
recovering	Healing, resting, or clearing stress
captured	Special war or intrigue state
retired	Inactive legacy state

XP and leveling

A practical leveling model:

```
xp_to_next_level(level) = 100 + floor(25 * level^2.2)
```

Task XP should be based on **risk, novelty, and consequence**, not raw repetition.

```
xp_gain = base_task_xp
         * difficulty_multiplier
         * success_multiplier
         * freshness_modifier
         * danger_modifier
```

Recommended `base_task_xp` ranges:

Activity	Base XP
Small speech in loyal city	8
Governing a city through one daily tick	10
Successful diplomacy action	15
Winning a battle as commander	25
Surviving a difficult intrigue mission	30
Resolving a nation-defining story event	40

Level-up rewards:

- +1 attribute point every level
- +1 skill perk every 2 levels
- trait roll every 4 levels or after major story events
- title/reputation unlocks at milestone levels

Player avatar policy

For the MVP, the player avatar should **not die permanently**. Severe failures should create injuries, scandals, exile, or temporary incapacitation. A full life-sim mortality/family system belongs in a later phase once the core agent loop works.

Event template format

The event system should be **data-driven** and support nation-, province-, city-, character-, and army-scoped events.

Recommended event JSON shape

```
{
  "id": "evt_border_port_strike",
  "version": 1,
  "scopeType": "province",
  "category": "military",
  "title": "Raiders in the Harbor",
  "summary": "A fast enemy flotilla has struck supply ships near the port.",
  "tags": ["war", "naval", "logistics"],
  "trigger": {
    "all": [
      { "path": "province.hasPort", "op": "eq", "value": true },
      { "path": "nation.atWar", "op": "eq", "value": true },
      { "path": "province.enemyNavalThreat", "op": "gte", "value": 1 }
    ]
  },
  "weightExpr": "10 + province.tradeValue * 0.4 + nation.unrest * 0.2",
  "cooldownHours": 72,
  "choices": [
    {
      "key": "fortify_port",
      "label": "Redirect engineers and fortify the harbor",
      "aiWeightExpr": "character.stewardship + 4",
      "effects": [
        { "type": "add_modifier", "scope": "province", "key":
"fortified_harbor", "durationDays": 14 },
        { "type": "change_resource", "scope": "nation", "resource": "money",
"delta": -120 }
      ]
    },
    {
      "key": "launch_pursuit",
      "label": "Send the local admiral to pursue",
      "aiWeightExpr": "character.martial + 6",
      "effects": [
        { "type": "spawn_followup_battle", "scope": "province", "battlePreset":
"coastal_skirmish" },
        { "type": "grant_xp", "scope": "character", "target":
"assignedCommander", "delta": 18 }
      ]
    },
    {
      "key": "issue_speech",
      "label": "Turn the raid into a patriotic rally",
      "aiWeightExpr": "character.charisma + 5",
      "effects": [
```

```

        { "type": "change_stat", "scope": "nation", "key": "legitimacy",
"delta": 3 },
        { "type": "create_post", "scope": "nation", "postTemplate":
"war_broadcast_port_strike" }
    ]
}
]
}

```

This format is deliberately close to how CK3/EU4/HOI4 content is conceptually authored—condition/effect driven—even though the exact runtime implementation here is original. 38

Seed data examples

Seed nation

```

{
  "name": "The Republic of Northreach",
  "adjective": "Northreachian",
  "governmentType": "Constitutional Republic",
  "currencyName": "Crownmark",
  "motto": "Steady Hands, Open Seas",
  "leaderTitle": "First Minister",
  "colorHex": "#355070",
  "startingStats": {
    "stability": 58,
    "legitimacy": 61,
    "influence": 20,
    "treasury": 1000,
    "researchPoints": 0
  }
}

```

Seed character

```

{
  "nationSlug": "northreach",
  "firstName": "Elara",
  "lastName": "Morne",
  "title": "First Minister",
  "isPlayerAvatar": true,
  "level": 1,
  "xp": 0,
  "martial": 4,
  "diplomacy": 8,

```

```
"stewardship": 6,  
"intrigue": 3,  
"charisma": 7,  
"traits": ["orator", "disciplined"]  
}
```

Seed unit template

```
{  
  "nationSlug": "northreach",  
  "name": "Line Marine Brigade",  
  "branch": "land",  
  "chassis": "infantry",  
  "role": "line",  
  "baseStats": {  
    "attack": 12,  
    "defense": 14,  
    "mobility": 8,  
    "siege": 4,  
    "vision": 5  
  },  
  "upkeep": {  
    "money": 8,  
    "food": 3,  
    "munitions": 2  
  },  
  "doctrineTags": ["coastal", "professional"]  
}
```

Seed event

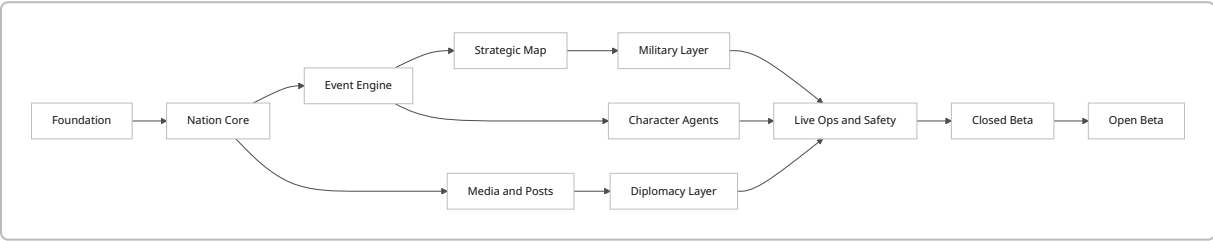
```
{  
  "id": "evt_capital_debate_housing",  
  "scopeType": "nation",  
  "category": "domestic",  
  "title": "The Housing Debate",  
  "summary": "Urban rents in the capital are becoming politically explosive.",  
  "choices": [  
    "subsidize_housing",  
    "free_market_reforms",  
    "ignore_for_now"  
  ]  
}
```

Roadmap and milestone flow

Recommended milestone sequence

Milestone	Main deliverables	Relative effort
Foundation	monorepo, auth, world bootstrap, shared types, admin seed scripts	Medium
Nation core	nation creation, profile tabs, law model, daily summary dashboard	Large
Media and posts	newsroom, story posts, image upload, video upload, visibility rules	Medium
Event engine	template parser, trigger evaluator, inbox UI, effects system, admin editor	Large
Strategic map	province graph, ownership, structures, resources, pathfinding, hover sheets	Large
Military layer	unit templates, production, armies, orders, battle resolver, battle reports	Large
Character agents	character roster, assignments, XP, traits, governors, commanders, speeches	Large
Diplomacy layer	treaties, embassies, friends, notifications, coalition tools	Medium
Live-ops and safety	moderation queue, rate limits, analytics, balancing tools, rollback scripts	Medium
Beta polish	onboarding, tutorials, balancing pass, AI helper nations/NPCs	Large

Milestone flow diagram



Suggested release scopes

Vertical slice

Ship first:

- account + nation creation
- nation profile
- event inbox
- one test world map
- one buildable city type
- one character assignment flow
- one army movement flow
- nation news post with image upload

That vertical slice is enough to prove the core thesis.

Closed alpha

Add:

- multiple province types
- 8–12 event templates
- 6–8 traits
- 8–10 unit templates
- treaties/friends
- battle reports
- moderator tools

Public beta

Add:

- video posts
- coalition tools
- AI-controlled background nations
- larger world
- richer local economies
- more narrative arcs
- better battle visualization

Open questions and limitations

A few important design choices remain open, and they are worth deciding explicitly before coding begins.

The first is **AI depth**. A “deep character agent system” can mean anything from score-based assignment heuristics to semi-autonomous long-horizon simulation. The safe recommendation is to begin with **goal-**

scored scheduled agents—strong enough to govern cities, lead armies, and perform missions—before attempting life-sim-grade household simulation.

The second is **battle granularity**. The plan above assumes **province-level positional warfare with summary battle resolution** rather than tactical battlefield micromanagement. That is the best fit for the browser and for the user’s nation-roleplay emphasis, but it should be affirmed as a hard scope boundary.

The third is **content moderation**. Because the user wants images and video attached to nation posts, moderation, abuse reporting, media scanning, and account safety should be designed in from the start, even if the first implementation is lightweight.

The fourth is **social topology**. This report intentionally minimizes regions because the user said they are not a primary interest. If later testing shows that players want stronger communal identity, the right expansion path is **coalitions, continental institutions, and optional public blocs**, not a return to deep region-governor gameplay.

The fifth is **source completeness for some browser-sim comparisons**. The strongest evidence in this report comes from official NationStates, Paradox, 2K, and framework documentation. Some Politics & War and eRepublik details were available most clearly through official search-result snippets rather than fully opened pages, so those comparisons are directionally useful but less granular than the NationStates/Paradox/Civ evidence base.

1 3 5 7 8 16 17 33 34 <https://www.nationstates.net/page%3Dfaq>
<https://www.nationstates.net/page%3Dfaq>

2 18 <https://vite.dev/guide/>
<https://vite.dev/guide/>

4 10 38 <https://www.paradoxinteractive.com/games/crusader-kings-iii/about>
<https://www.paradoxinteractive.com/games/crusader-kings-iii/about>

6 <https://www.nationstates.net/page%3Ddispatches/category%3D1>
<https://www.nationstates.net/page%3Ddispatches/category%3D1>

9 36 https://store.steampowered.com/app/289070/Sid_Meiers_Civilization_VI/
https://store.steampowered.com/app/289070/Sid_Meiers_Civilization_VI/

11 <https://www.paradoxinteractive.com/games/hearts-of-iron-iv/about>
<https://www.paradoxinteractive.com/games/hearts-of-iron-iv/about>

12 <https://www.paradoxinteractive.com/games/europa-universalis-iv>
<https://www.paradoxinteractive.com/games/europa-universalis-iv>

13 <https://politicsandwar.com/index.php>
<https://politicsandwar.com/index.php>

14 <https://erepublik.com/>
<https://erepublik.com/>

15 <https://playsupremacy.com/>
<https://playsupremacy.com/>

19 <https://nextjs.org/docs>

<https://nextjs.org/docs>

20 <https://pixijs.com/>

<https://pixijs.com/>

21 <https://phaser.io/>

<https://phaser.io/>

22 https://docs.godotengine.org/en/latest/tutorials/export/exporting_for_web.html

https://docs.godotengine.org/en/latest/tutorials/export/exporting_for_web.html

23 <https://docs.unity3d.com/6000.4/Documentation/Manual/webgl.html>

<https://docs.unity3d.com/6000.4/Documentation/Manual/webgl.html>

24 <https://docs.nestjs.com/>

<https://docs.nestjs.com/>

25 <https://docs.colyseus.io/>

<https://docs.colyseus.io/>

26 <https://socket.io/docs/v4/>

<https://socket.io/docs/v4/>

27 <https://www.postgresql.org/docs/current/tutorial-transactions.html>

<https://www.postgresql.org/docs/current/tutorial-transactions.html>

28 <https://www.prisma.io/docs/orm>

<https://www.prisma.io/docs/orm>

29 <https://doc.mapeditor.org/en/stable/reference/json-map-format/>

<https://doc.mapeditor.org/en/stable/reference/json-map-format/>

30 <https://www.cloudflare.com/developer-platform/products/r2/>

<https://www.cloudflare.com/developer-platform/products/r2/>

31 <https://docs.aws.amazon.com/AmazonS3/latest/userguide/Welcome.html>

<https://docs.aws.amazon.com/AmazonS3/latest/userguide/Welcome.html>

32 <https://pixijs.com/8.x/guides/getting-started/intro>

<https://pixijs.com/8.x/guides/getting-started/intro>

35 <https://www.nationstates.net/page%3Ddispatch/id%3D1381642>

<https://www.nationstates.net/page%3Ddispatch/id%3D1381642>

37 https://forumcontent.paradoxplaza.com/public/paradox/banners/HoI_IV_Strategy_Guide.pdf

https://forumcontent.paradoxplaza.com/public/paradox/banners/HoI_IV_Strategy_Guide.pdf